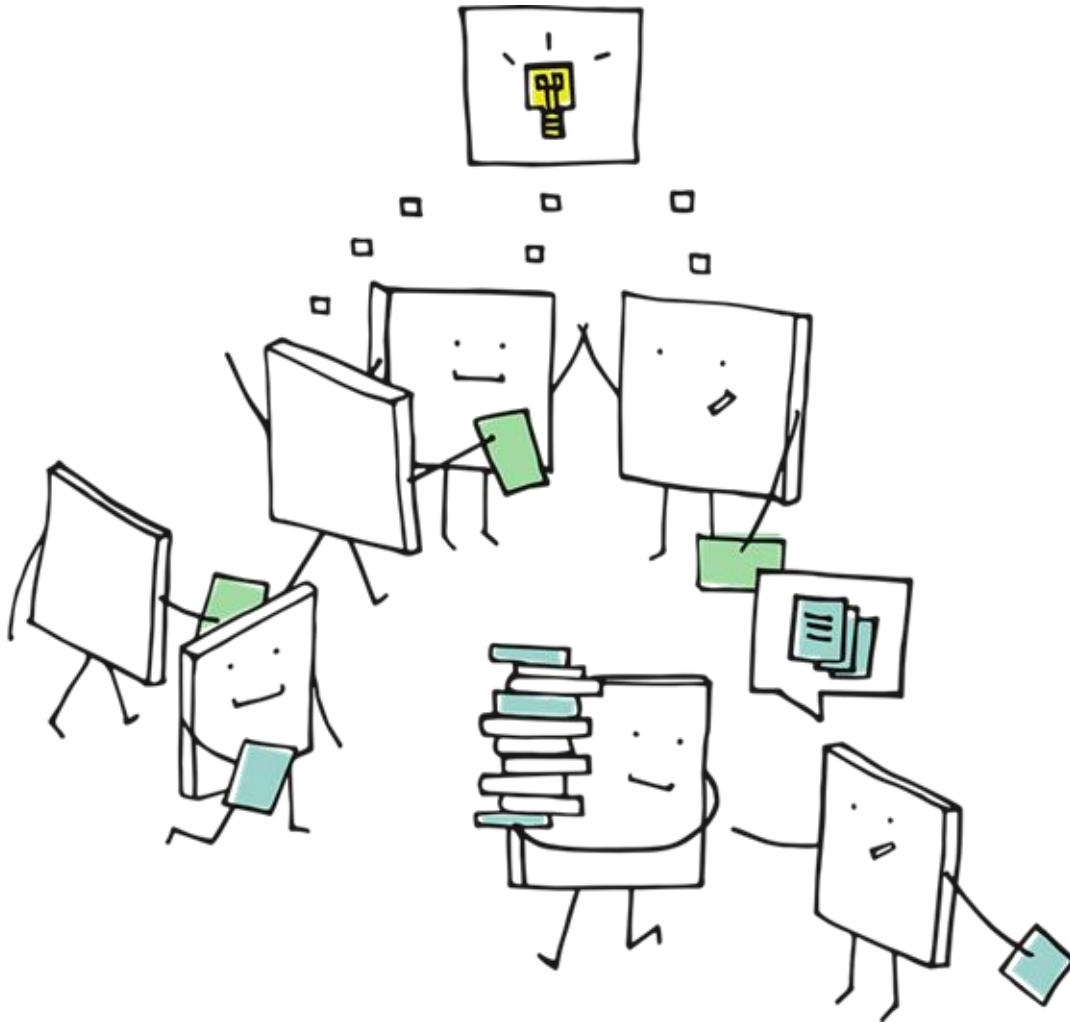




Generative AI *using* LLM

1. What is Generative AI

2. What is LLM?

3. Using LLMs - Zero Shot Learning

4. Using LLMs - Fine Tuning LLMs

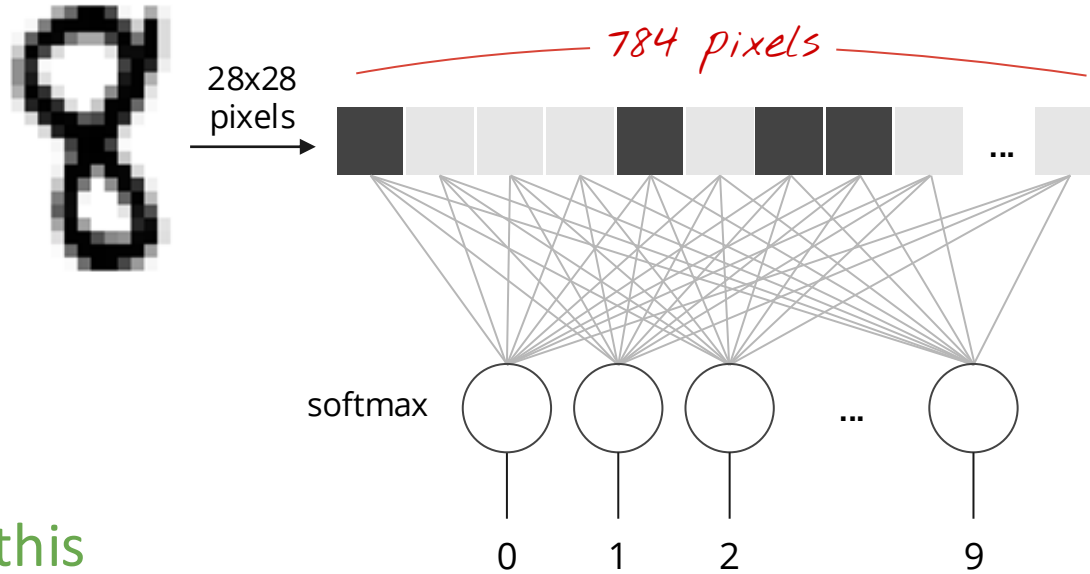
5. LangChain

6. Diffusion Models



What is the price of
a house?

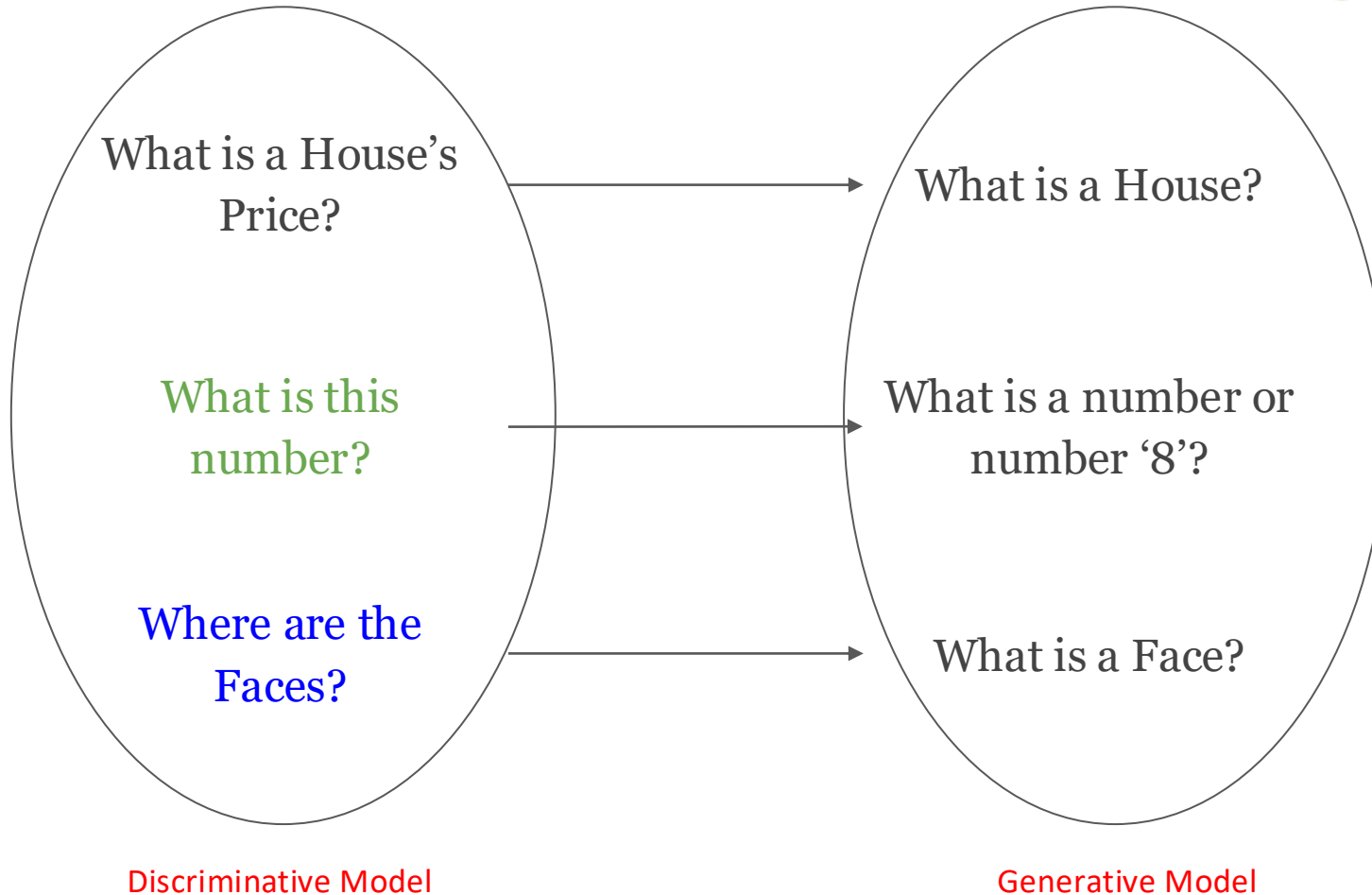
So far, we have build models which predict things like...



What is this
number?

Where are the Faces?





#	Text to Label Task	Input	Classes or Categories
1	Customer Support ticket classification	Customer support tickets or emails	Billing Issue, Technical Support, Product Inquiry, etc.
2	Sentiment Analysis for Product Reviews	Product reviews or comments	Positive, Negative or Neutral
3	Topic Categorization for News Articles	New Articles	Politics, Sports, Entertainment, etc
4	Resume Screening	Job applications	Suitable for Interview, Not Suitable, etc.
5	Social Media Post Classification	Social media posts	News, Events, Personal Updates, etc.
6	Legal Document Categorization	Legal documents	Contract, Complaint, Agreement, etc.
7	Product Categorization for eCommerce	Product descriptions	Electronics, Clothing, Home Appliances, etc.
8	Medical Record Classification	Electronic Health Records	Diagnosis, Medication, Treatment Plan, etc.

Text to **Label** examples
(Discriminative)

#	Text to Text Task	Input	Classes or Categories
1	Customer Support Automation	Customer inquiries or complaints in text form	Automated responses or suggested solutions generated in response to customer queries.
2	Email Autoresponders	Incoming emails	Automatic email responses generated based on predefined rules or machine learning algorithms.
3	Text Summarization	Long documents, articles, or text passages.	Summarized versions of the input text, condensing the key information
4	Content Generation	Prompts or specifications for content creation	Generated articles, stories, or creative pieces in response to the input
5	Code Generation	Descriptions or requirements for a program	Automatically generated code based on the input specifications
6	Social Media Posts	Drafts or prompts for social media posts	Published social media content

Text to **Text** examples (Generative)



What is **Generative AI**?

What is Generative AI?

Definition

Generative AI refers to models that can create new content — text, images, audio, code, video — by learning patterns from training data and generating statistically likely continuations or transformations.

Core Idea

Unlike discriminative models (which classify inputs), generative models learn the joint distribution $P(X)$ or $P(X|Y)$ and sample from it — producing novel, coherent outputs.

Key Modalities

Text (GPT, LLaMA)
Images (DALL·E, Stable Diffusion)
Audio (Whisper, MusicGen)
Code (Codex, CodeLLaMA)
Video (Sora, Runway)

Scale (2025)

ChatGPT: 300M+ weekly users
Gemini: integrated in 1B+ devices
LLM market: \$40B → \$1.3T by 2032
Tokens processed daily: trillions



What are Large Language Models (**LLMs**)?

What is a Large Language Model (LLM)?

A Large Language Model is a neural network trained on massive text corpora to predict the next token — and through that simple objective, emergent capabilities arise.

1. Scale

Billions–Trillions of parameters
Trillions of training tokens
Thousands of GPUs for months

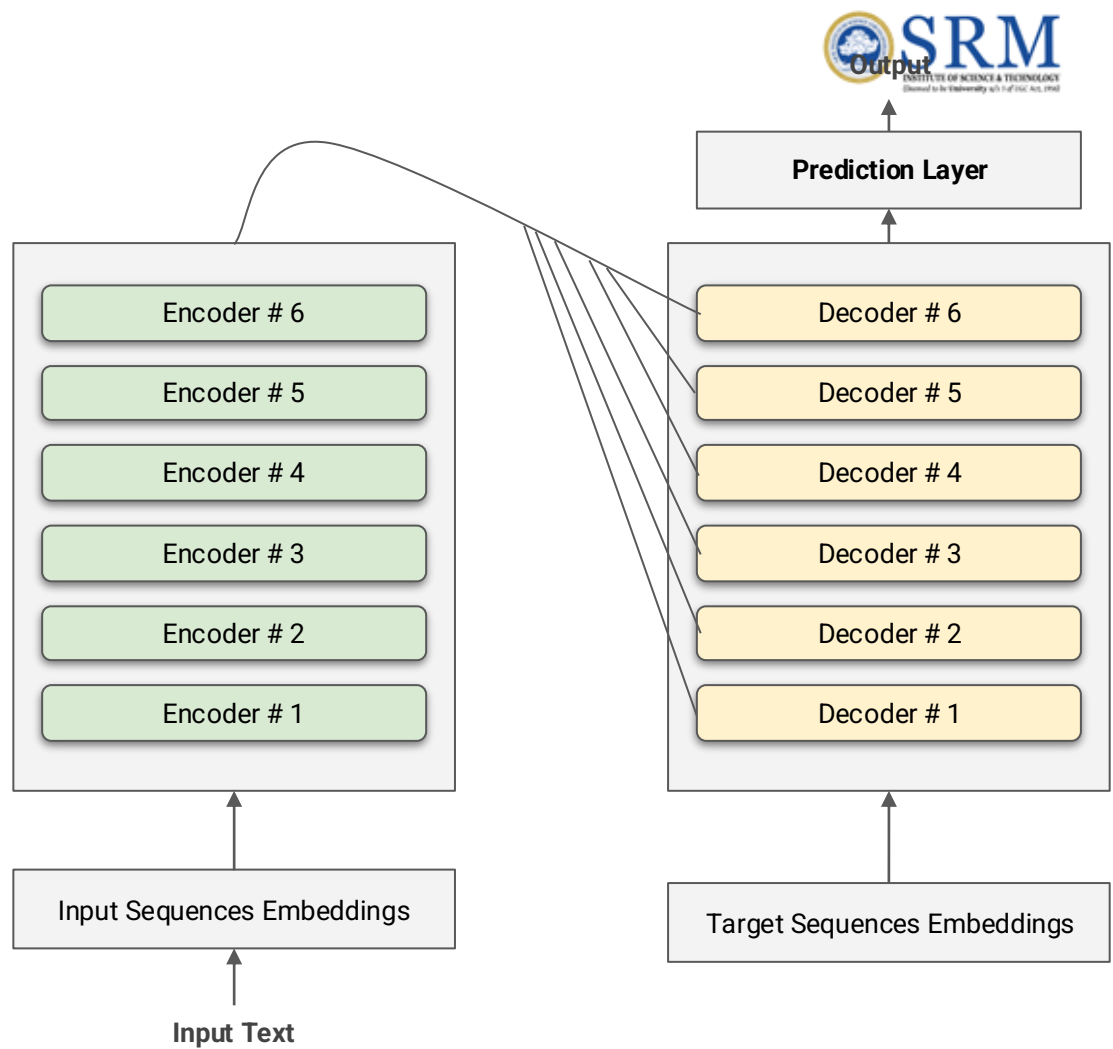
2. Architecture

Transformer-based
Self-attention mechanism
Decoder-only or Enc-Dec

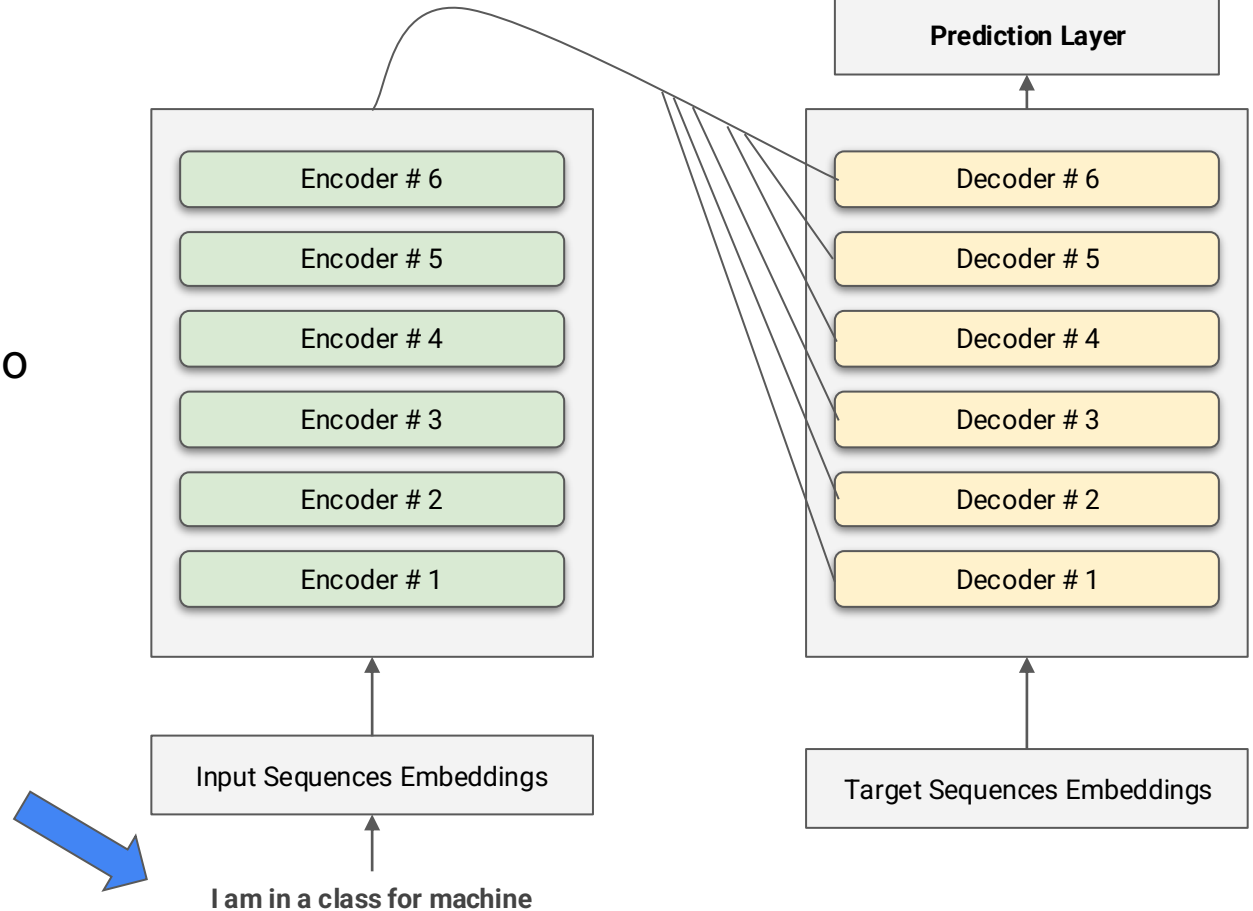
3. Emergence

Chain-of-thought reasoning
Few-shot learning
Instruction following

LLMs typically use
Transformer architecture



LLMs are usually trained to predict **NEXT WORD**



How LLMs works



What is a Token?

A token is the atomic unit of text an LLM processes.
"tokenization" = 3 tokens. A token $\approx$ 0.75 words.

Example: "Hello, world!" $\rightarrow$ ["Hello", ",", " world", "!"]



What is an Embedding?

An embedding is a dense vector (e.g. 4096 dimensions) that represents a token in semantic space. Similar meanings $\rightarrow$ similar vectors. This is how LLMs understand context.



Why This Matters

Context window = max tokens the model can "see" at once.

GPT-4: 128K tokens | Gemini 1.5: 1M tokens | Claude 3.5: 200K tokens

More context = longer conversations, bigger codebases, larger documents in memory.

How Large Language Models (LLMs) are Made?⁴

1. Collect vast amounts of data (**training data**)
2. Transform data into tokens (parts of words) → trained on 500 billion tokens!
3. Unsupervised learning: training the model (weeks to months). It creates a huge mathematical representation of human language.
4. **Fine-tuning**: Expose it to new tasks and give it more guidelines
5. **Reinforcement learning with human feedback (RLHF)**: Use human reviewers to rate responses to improve quality. The model learns from those ratings.
6. **Deploy to the public**: GPT 3.5 → “a research preview.” Learn from more people using it so it can be improved.

Pre-training — How LLMs Learn

Objective: Predict the next token given all previous tokens → Causal Language Modeling (CLM)

Training Data

CommonCrawl, Wikipedia, Books, GitHub, ArXiv — trillions of tokens scraped and filtered

Compute

GPT-4 training: est. \$100M+, months on thousands of A100/H100 GPUs

Objective

Cross-entropy loss: minimize $-\log P(\text{next_token} \mid \text{context})$

Result

Model implicitly learns grammar, facts, reasoning, code, math — all from next-token prediction

RLHF — Making LLMs Helpful & Safe

Reinforcement Learning from Human Feedback (RLHF) aligns pre-trained LLMs with human values, preferences, and safety constraints.

1

Supervised Fine-Tuning (SFT)

Human annotators write ideal responses. Model fine-tuned on these demonstrations.

2

Reward Model Training

Humans rank multiple model outputs. A reward model learns to predict human preference scores.

3

PPO Optimization

The LLM is updated via Proximal Policy Optimization to maximize reward model scores — constrained to not drift too far from SFT baseline.

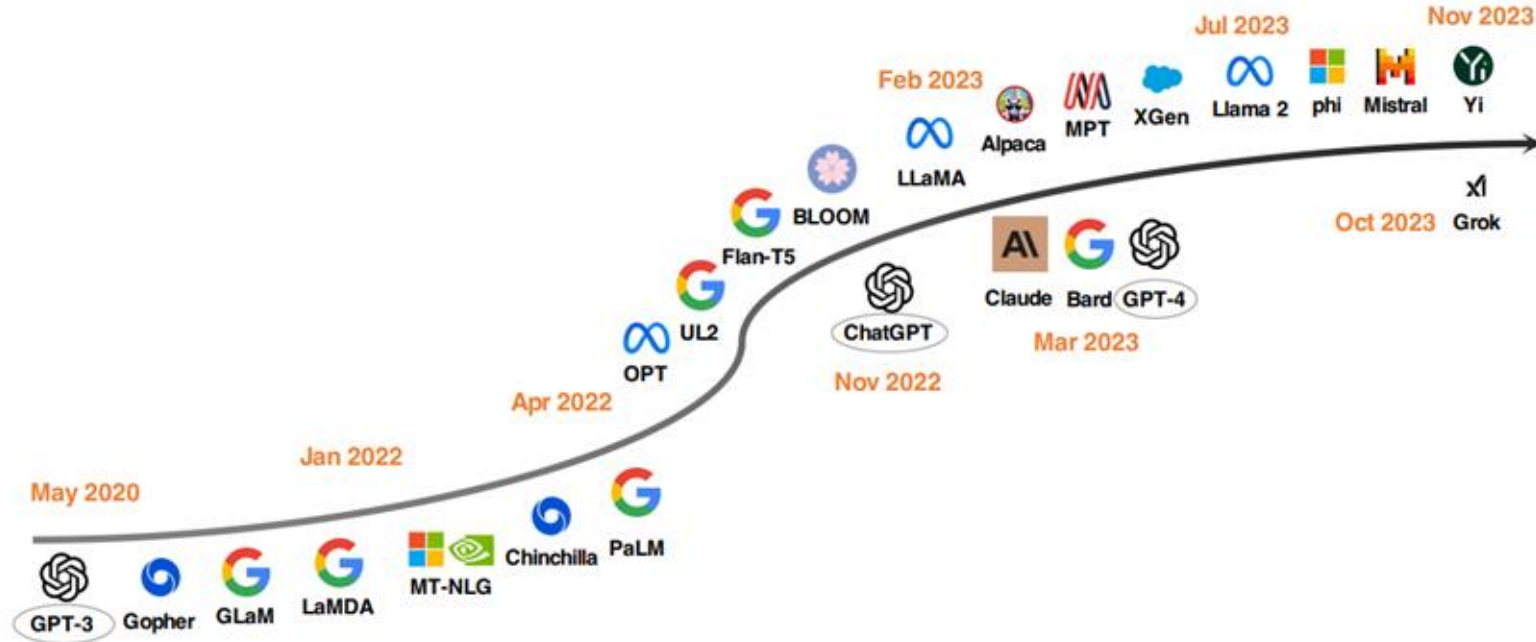
LLMs are trained using **HUGE DATA** in **Self-Supervised** Learning



Where do we get huge datasets?

Check out the list of datasets available (not a comprehensive list)

<https://github.com/Zjh-819/LLMDataHub>



A brief history of LLMs

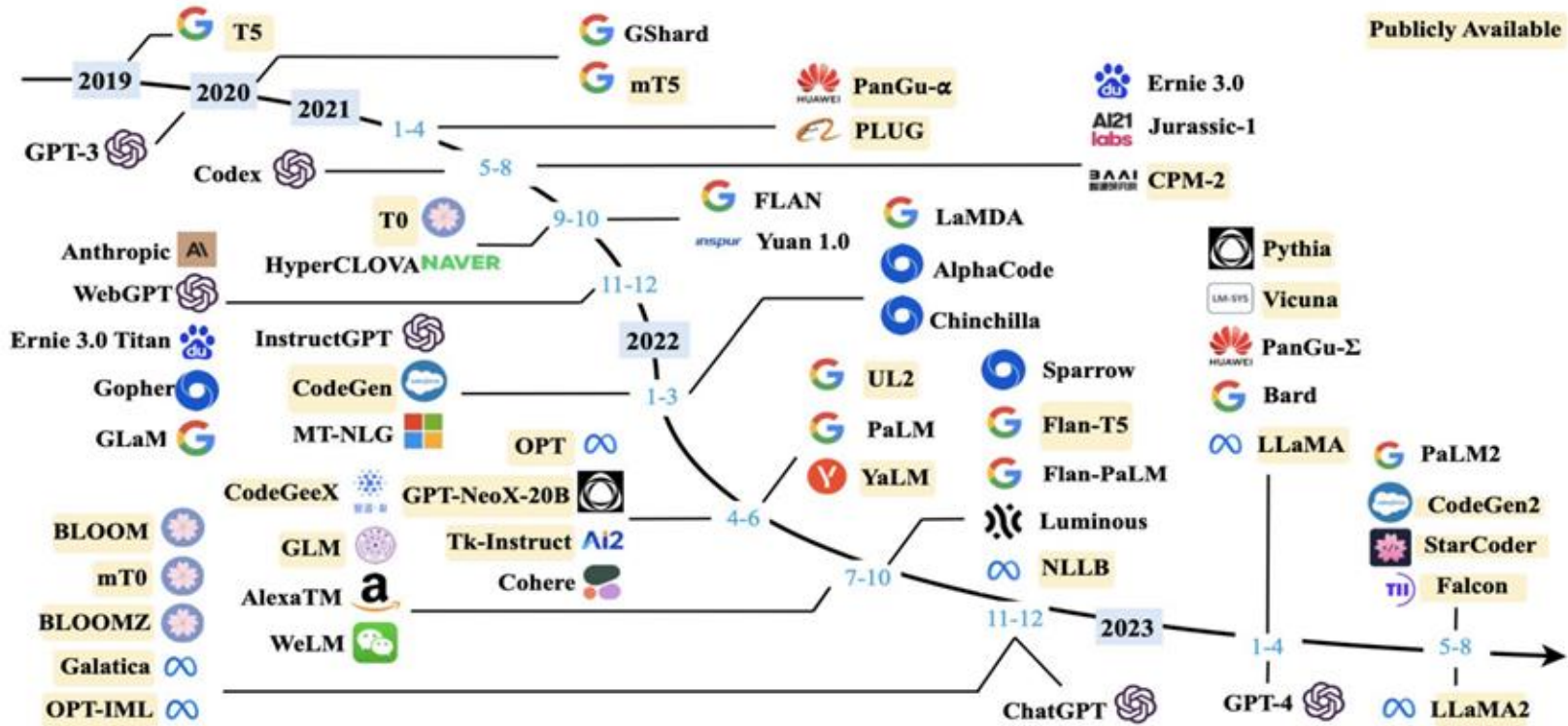
Source: <https://arxiv.org/pdf/2311.16989.pdf>

LLMs are usually quite **Big** in number of parameters (Weights & Bias)

Models	Publication Venue	License Type	Model Creators	Purpose	No. of Params	Commercial Use	Steps Trained	Data/ Tokens
T5 [10]	JMLR'20	Apache-2.0	Google	General	11B	✓	1M	1T
GPT-3 [6]	NeurIPS'20	-	OpenAI	General	175B	×	-	300B
mT5 [11]	NAACL'21	Apache-2.0	Google	General	13B	✓	1M	1T
PanGu- α [105]	arXiv'21	Apache-2.0	Huawei	General	200B	✓	260k	1.1TB
CPM-2 [12]	AI Open'21	MIT	Tsinghua	General	198B	✓	1M	2.6TB
Codex [132]	arXiv'21	-	OpenAI	Coding	12B	×	-	100B
ERNIE 3.0 [108]	arXiv'21	-	Baidu	General	10B	×	120k*	375B
Jurassic-1 [110]	White-Paper'21	Apache-2.0	AI21	General	178B	✓	-	300B
HyperCLOVA [112]	EMNLP'21	-	Naver	General	82B	×	-	300B
Yuan 1.0 [113]	arXiv'21	Apache-2.0	-	General	245B	✓	26k*	180B
Gopher [114]	arXiv'21	-	Google	General	280B	×	-	300B
ERNIE 3.0 Titan [35]	arXiv'21	-	Baidu	General	260B	×	-	300B
GPT-NeoX-20B [274]	BigScience'22	Apache-2.0	EleutherAI	General	20B	✓	150k	825GB
OPT [14]	arXiv'22	MIT	Meta	General	175B	✓	150k	180B
BLOOM [13]	arXiv'22	RAIL-1.0	BigScience	General	176B	✓	-	366B
Galactica [139]	arXiv'22	Apache-2.0	Meta	Science	120B	×	225k	106B
GLaM [119]	ICML'22	-	Google	General	1.2T	×	600k*	600B
LaMDA [141]	arXiv'22	-	Google	Dialog	137B	×	3M	2.81T
MT-NLG [115]	arXiv'22	Apache-v2.0	MS.+Nvidia	General	530B	×	-	270B
AlphaCode [133]	Science'22	Apache-v2.0	Google	Coding	41B	✓	205k	967B
Chinchilla [122]	arXiv'22	-	Google	General	70B	×	-	1.4T
PaLM [15]	arXiv'22	-	Google	General	540B	×	255k	780B
AlexaTM [123]	arXiv'22	Apache v2.0	Amazon	General	20B	×	500k	1.1T
U-PaLM [125]	arXiv'22	-	Google	General	540B	×	20k	-
UL2 [89]	ICLR'23	Apache-2.0	Google	General	20B	✓	2M	1T
GLM [33]	ICLR'23	Apache-2.0	Multiple	General	130B	×	-	400B
CodeGen [131]	ICLR'23	Apache-2.0	Salesforce	Coding	16B	✓	650k	577B
LLaMA [127]	arXiv'23	-	Meta	General	65B	×	350k	1.4T
PanGu Σ [130]	arXiv'23	-	Huawei	General	1.085T	×	-	329B
BloombergGPT [142]	arXiv'23	-	Bloomberg	Finance	50B	×	139k	569B
Xuan Yuan 2.0 [144]	arXiv'23	RAIL-1.0	Du Xiaoman	Finance	176B	✓	-	366B
CodeT5+ [34]	arXiv'23	BSD-3	Salesforce	Coding	16B	✓	110k	51.5B
StarCoder [138]	arXiv'23	OpenRAIL-M	BigCode	Coding	15.5B	✓	250k	1T
LLaMA-2 [21]	arXiv'23	LLaMA-2.0	Meta	General	70B	✓	500k	2T
PaLM-2 [124]	arXiv'23	-	Google	General	-	×	-	-

Source

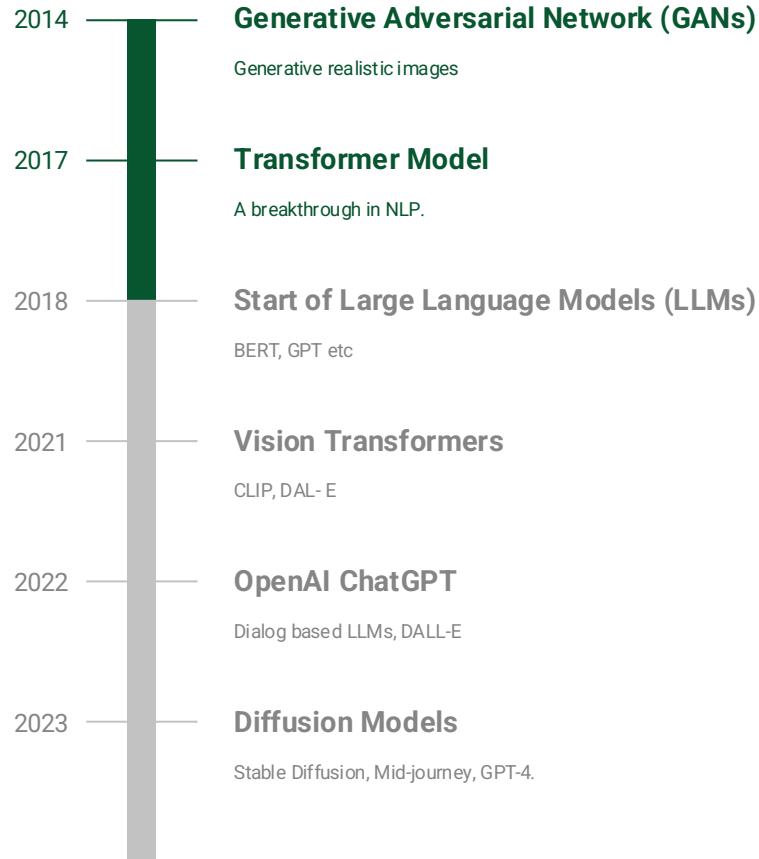
<https://arxiv.org/pdf/2307.06435.pdf>



LLMs Chronology >10B



How do we allow Models to be
Generative?



A timeline of Generative AI

The GPT Lineage — OpenAI's Scaling Story



	GPT-1 (2018)	117M params · 1B tokens	Unsupervised pre-training + fine-tuning showed viability
	GPT-2 (2019)	1.5B params · 40B tokens	"Too dangerous to release" — generated coherent long-form text
	GPT-3 (2020)	175B params · 300B tokens	Few-shot learning without fine-tuning; in-context learning emerged
	InstructGPT (2022)	~175B params · +RLHF tokens	RLHF introduced — aligned to follow instructions safely
	GPT-4 (2023)	~1.8T (est) params · 13T tokens	Multimodal, MMLU top-performer; passes bar exam
	GPT-4o (2024)	N/A params · N/A tokens	Omni-model: native speech+vision+text, 2× faster
	o1 / o3 (2025)	N/A params · N/A tokens	Chain-of-thought reasoning; competitive math/code scores

The Open-Source LLM Explosion (2023–2026)

- 🐘 LLaMA 1 (Feb 2023) — Meta released weights; enabled academic/community fine-tuning
 - 🐘 LLaMA 2 (Jul 2023) — Commercializable license; Llama 2-Chat became ChatGPT competitor
 - 🐘 LLaMA 3.1 (2024) — 405B model rivals GPT-4 on benchmarks
 - 🦋 Mistral 7B — Outperforms Llama 2-13B despite 2× fewer params (efficient arch)
 - 🌐 Mixtral 8x7B — Mixture-of-Experts; 2 experts active per token → 12.9B effective
- 😊 Hugging Face ecosystem — 500K+ model repos; democratizing AI
 - ⚙️ PEFT / LoRA / QLoRA — Fine-tune 7B model on a single GPU
 - ⚡ Ollama / LMStudio — Run LLMs locally on laptop; privacy-first
 - 📦 GGUF quantization — 4-bit models; GPT-3 level on MacBook
 - 🌍 Impact — LLM development no longer only for billion-dollar labs

2024–2026: Multimodal, Agents & the Agentic Turn

Multimodal LLMs

GPT-4V, Gemini 1.5, Claude 3
Text + Images + Audio + Video natively.
"See" a PDF, hear a question, respond
across modalities.

Long Context

Gemini 1.5 Pro: 1M tokens
= 5 full novels in context
Code repos, legal docs, full videos

AI Agents

LLMs that plan, use tools, browse web,
write+execute code, call APIs. AutoGPT
→ CrewAI → OpenAI Swarms →
Anthropic MCP

SLMs / Edge

Phi-3-mini (3.8B) matches GPT-3.5
On-device models in phones
Apple Intelligence, Gemini Nano

Reasoning Models

o1/o3 "think before answering"
Internal scratchpad
Competitive math olympiad scores

Vibe Coding

Claude Code, Cursor, GitHub Copilot
LLMs write entire codebases from
prompts
"Programmer" role redefined

Traditional DNN vs LLM — High-Level Contrast

Traditional DNN

- Task-specific (e.g. classify cats)
- Trained per task from scratch
- Fixed input/output format
- Supervised labels required
- Hundreds of MBs
- No language understanding
- No generalization to new tasks

Large Language Model

- General-purpose across tasks
- Pre-trained once, adapt via prompts
- Natural language in/out
- Self-supervised (next token)
- Hundreds of GBs / TBs
- Deep language understanding
- Zero-shot to new tasks

Training Paradigm: Supervised vs Self-Supervised



Traditional: Supervised Learning

Requires labeled data: (input, label) pairs

Example: 10,000 cat/dog images labeled by humans

Limitation: Expensive labels; can't scale infinitely

Model: learns $f(x) \rightarrow y$ for THAT task only



LLM: Self-Supervised Learning

Labels ARE the data. Next token = label.

"The cat sat on the [MASK]" $\rightarrow$ "mat"

Uses all text on the internet as training signal

Model: learns rich representations of language



Key Insight — The Scaling Hypothesis

With enough data + compute + parameters, next-token prediction forces the model to internalize all knowledge needed to predict language — including facts, reasoning, and world models. This is why "just predicting the next word" produces such capable systems.

Architecture: CNN/RNN vs Transformer

Feature	CNN	RNN / LSTM	Transformer (LLM)
Core Op	Convolution	Sequential recurrence	Self-attention
Long-range	Limited by kernel	Vanishing gradient	Full context access
Parallelism	High (spatial)	Low (sequential)	High (attention heads)
Best For	Images, local patterns	Time-series, short seq	Language, long sequences
Scale	Millions	Millions	Billions–Trillions
In-context	No	No	Yes — via prompts

Task Generalization — The Key LLM Superpower

A traditional model trained for sentiment analysis CANNOT translate text. An LLM trained on language can do BOTH — and hundreds of other tasks — with just a prompt change.

✓ Sentiment Analysis

✓ Translation

✓ Summarization

✓ Code Generation

✓ Question Answering

✓ Data Extraction

✓ Creative Writing

✓ Classification

✓ Reasoning

All achieved with a single LLM — no retraining, just prompting

Emergent Capabilities — What Scale Unlocks

Emergent abilities are capabilities that appear suddenly at certain scale thresholds — not present in smaller models.

Chain-of-Thought

Step-by-step reasoning emerges at ~100B params

100B+

Arithmetic

Multi-digit arithmetic without calculators

50B+

Code Execution

Write + debug code from natural language specs

20B+

Instruction Follow

Follows complex multi-step instructions reliably

7B+

Theory of Mind

Models others' beliefs and intentions

175B+

Prompt Generation

Need good prompts to get
good responses!

CLEAR Framework⁶

Concise
Logical
Explicit
Adaptive
Reflective



1. Concise

- Be specific
- Use simple language
- Prioritize critical information

Concise

Logical

Explicit

Adaptive

Reflective

Instead of “What causes society to change?” go with,

Example prompt:

List three of the most significant social factors during the industrial revolution?

Example Prompt:



What causes society to change?



Society changes for a variety of complex and interconnected reasons. These changes can be driven by a combination of social, economic, political, technological, cultural, and environmental factors. Here are some key factors that contribute to societal change:

1. **Technological Advances:** Technological innovations often lead to significant societal changes. The invention of the printing press, for example, played a crucial role in the spread of knowledge and the Protestant Reformation. Similarly, the internet and digital technologies have transformed how people communicate, work, and access information in the modern era.
2. **Economic Forces:** Economic factors such as industrialization, globalization, and changes in the job market can drive societal change. The transition from agrarian economies to industrial economies, for instance, brought about urbanization and shifts in social structures.

VS.



List three of the most significant social factors during the industrial revolution?



The Industrial Revolution, which occurred in the late 18th and early 19th centuries, brought about significant social changes. Three of the most significant social factors during this period were:

1. **Urbanization:** The Industrial Revolution led to a massive migration of people from rural areas to urban centers in search of employment in factories and industries. This rapid urbanization resulted in the growth of cities and the development of a new urban working class. Urban areas became crowded and often lacked adequate housing, sanitation, and infrastructure, leading to poor living conditions for many.
2. **Labor Conditions:** The working conditions in factories and mines during the Industrial Revolution were often harsh and dangerous. Long working hours, low wages, and the absence of workers' rights and safety regulations contributed to the exploitation of laborers. Child labor was widespread, with

2. Logical

- Structure info in order (logical flow) → Conversational!
- Establish context and relationships
- Avoid too many instructions in a single prompt⁷

[write them out one at a time]

Example prompts:

- Tell me about the history of the Eiffel Tower.
- Were there any major controversies surrounding the project?
- What major construction materials were used?
- Tell me about its design and designers
- Explain its significance
- Let's talk about the budget

3. Explicit

- **Define instructions**
- Set reading levels, output formats
- Assign a role for ChatGPT to play

Example prompt:

Could you please explain the following passage?

"The value of the modulus of rigidity shows the resistance of a given material to shear deformation. Engineers consider the value of shear modulus when selecting materials for shafts or rods that are subjected to twisting torques."

3. Explicit

- Define instructions
- **Set reading levels, output formats**
- Assign a role for ChatGPT to play

Concise
Logical
Explicit
Adaptive
Reflective

Instead of “How to apply for the Marie Curie fellowship,” go with something like,⁸

Example Prompt:

Please tell me how to apply for a Marie Curie fellowship? Please limit your answer to five points and tell it to me like I'm a 10-year-old.

Other Output Formats:⁹

- Table
- Bulleted list
- Numbered list
- Summary
- Code
- Story/poem
- Dialogue
- Q&A
- Emojis ... and more!

3. Explicit

Concise
Logical
Explicit
Adaptive
Reflective

- Define instructions
- Set reading levels, output formats
- **Assign a role for ChatGPT to play**

Example Prompt 1:

You are an undergraduate science student.
Describe the Krebs cycle in simple terms.

Example Prompt 2:

Imagine you're a public relations representative dealing with a social media crisis.¹⁰

Draft a press release that acknowledges the issue, explains what actions your company is taking, and reassures the public.

Example Prompt:

KU

You are an undergraduate science student. Describe the Krebs cycle in simple terms.



Sure, I'd be happy to describe the Krebs cycle in simple terms!



The Krebs cycle, also known as the citric acid cycle or the TCA cycle (Tricarboxylic Acid Cycle), is a crucial part of how our cells produce energy. Imagine it as a bit like a power plant for your cells.

Here's how it works:

1. **Fuel In:** The cycle starts when a molecule called acetyl-CoA, which comes from the breakdown of food molecules like carbohydrates, fats, and proteins, enters the cycle.
2. **Molecule Shuffle:** Acetyl-CoA combines with a molecule called oxaloacetate to kickstart the cycle. This combination forms a molecule called citric acid.
3. **Chemical Transformations:** As citric acid goes through a series of chemical reactions, it releases energy and transfers electrons to carrier molecules, like NADH and FADH₂. These molecules are like little energy shuttles.
4. **Carbon Dioxide Out:** During these reactions, carbon dioxide is produced as a waste

4. Adaptive

- Be flexible (rephrase and restructure)
- Try different approaches (be more creative with prompts)

Example Prompt:

What are the ethical concerns with using ChatGPT?

Concise
Logical
Explicit
Adaptive
Reflective

4. Adaptive

- Be flexible (rephrase and restructure)
- Try different approaches (be more creative with prompts)

Example Prompt:

What are the ethical concerns with using ChatGPT?

Follow-up Prompt:

Can you provide 3 more examples?

5. Reflective

ChatGPT will never ever EVER be wrong!*

*** Well, actually... that's not true!**

- Carefully evaluate AI responses
- Identify areas for improvement (it takes time)
- Use insights to further refine strategies for engagement

Concise
Logical
Explicit
Adaptive
Reflective

Strengths

Writing	Learning/Studying
● Generate ideas ● Brainstorm research topics ● Change citation styles ● Editing and proofreading ● Suggestions for databases to find literature on your topic "Correct the grammar: [paragraph]" "Identify the gaps in the literature on [topic]"	● Can explain complex concepts ● Summarize information ● Practice second languages "Can you explain x in simple terms?" "How about using an analogy to describe it?" "Please summarize [insert book name] by [author] and provide the top 5 important themes."

Why Prompting Matters

10×

Output quality difference between a vague and a well-engineered prompt

\$0

Cost to improve results with better prompting vs fine-tuning

~95%

Tasks solvable with prompting alone; no retraining needed

- Prompting is the new programming — it replaces writing code for many automation tasks
- Bad prompt: "Summarize this" → vague, inconsistent output every time
- Good prompt: "Summarize this contract in 5 bullet points. Focus on obligations, deadlines, and payment terms. Output as JSON." → reliable, structured, usable
- Prompt engineering is the highest-ROI skill in AI adoption — costs nothing, yields everything
- Tools like LangChain, DSPy, and PromptFlow exist entirely to manage and optimize prompts at scale

Zero-Shot Prompting

Ask the model to perform a task without any examples. The model relies purely on pre-trained knowledge.

EXAMPLE

Classify the sentiment of this review as Positive, Neutral, or Negative:

"The product arrived quickly but the quality was disappointing."

Model Output: Negative

- Best for: common, well-defined tasks the model has seen during training
- Works well with large models (GPT-4, Claude 3, Gemini 1.5)
- Can fail on niche, domain-specific, or ambiguous tasks

Few-Shot Prompting

Include 2–5 examples of desired input→output pairs before your actual query. Dramatically improves accuracy.

```
Review: "Fast shipping, great product!" → Positive
```

```
Review: "It broke after 2 days." → Negative
```

```
Review: "Not bad, works as described." → Neutral
```

```
Now classify:
```

```
Review: "Came late but works perfectly." → ?
```

Output: Positive (model learned the pattern from examples)

Chain-of-Thought (CoT) Prompting

Instruct the model to reason step by step before giving its final answer. Dramatically improves multi-step reasoning.

Without CoT

Q: A store had 24 apples. They sold $\frac{1}{3}$ and then got 15 more. How many now?

Model: 24

Wrong! (correct = 31)

With CoT ("Think step by step")

Step 1: Start with 24 apples.

Step 2: Sold $\frac{1}{3} \rightarrow 24 \times \frac{1}{3} = 8$

Step 3: Remaining = $24 - 8 = 16$

Step 4: Received 15 more

Step 5: $16 + 15 = 31$

Answer: 31

Model Name	Accuracy	Source
ChatGPT (Zero Shot)	73	Link
ChatGPT (10 Examples - Few Shot)	80.8	Link
ChatGPT (20 Examples - Few Shot)	82.8	Link
ChatGPT (30 Examples - Few Shot)	83.0	Link
Fine Tuned BERT	84.0	

Text Classification

Twitter US Airline Sentiment Analysis

Role & Persona Prompting

Assigning a role to the model dramatically shifts its tone, depth, vocabulary, and priorities.



Doctor

"You are a licensed physician. Explain the risks of mixing ibuprofen and alcohol."

→ *Clinical, cautious, cites guidelines*



Teacher

"You are a 5th-grade science teacher. Explain how black holes work."

→ *Simple analogies, no jargon, engaging*



CFO

"You are a CFO reviewing a startup pitch deck. Give brutally honest financial feedback."

→ *Financial metrics focused, direct, critical*



Hacker

"You are a senior red-team security engineer. Review this code for vulnerabilities."

→ *Finds edge cases, CVE references, adversarial thinking*

Prompt Templates, Systems & Engineering Patterns

-  Template Pattern — Parameterize reusable prompts: "Summarize {{document}} in {{n}} bullets targeting {{audience}}"
 -  Chained Prompts — Break complex tasks into a pipeline: extract → analyze → format → validate
 -  Constitutional Prompting — Add constraints/rules: "Never reveal personal data; always cite sources"
 -  Self-Consistency — Run same prompt N times, take majority vote for reliability
-  ReAct (Reason + Act) — Model alternates between reasoning and tool calls: think, act, observe, repeat
 -  Structured Output — Force JSON/XML schema: "Respond ONLY with JSON matching this schema: {name, score, reason}"
 -  Self-Critique — "Review your previous answer and identify any errors or missing info"
 -  Meta-Prompting — Ask the model to write a better prompt for itself



Limitations

What are the Problems?

Limited Knowledge

Data until September 2021, not replicable

Bias

Misinformation

Ethical Concerns



Misinformation

- Sometimes ChatGPT makes stuff up, called **hallucinations**.
- Remember: based on the **probability** of what the next word should be
- Based on AI model's inherent biases and training dataset limitations
- "These hallucinations are particularly problematic in domains that require multi-step reasoning since a single logical error is enough to derail a much larger solution."¹¹

Examples of Hallucinations

User input:¹²

"When did Leonardo da Vinci paint the Mona Lisa?"

AI-generated response:

"Leonardo da Vinci painted the Mona Lisa in 1815."

(Incorrect: The Mona Lisa was painted between 1503 and 1506, or perhaps continuing until 1517.)



Bias

- Bias exists in the training data itself (sexism, racism, homophobia, ableism, etc.)
- Western, English focused (developed in the US)
- Not enough information or nuance for certain types of content in the dataset
- **Only humans** have the skillset to think critically, conduct nuanced research, and fact-check!

Ethical Concerns

BUSINESS • TECHNOLOGY

Exclusive: OpenAI Used Kenyan Workers on Less Than \$2 Per Hour to Make ChatGPT Less Toxic

- Copyright
- Low wages to humans training ChatGPT¹³
- Dependency on AI
- What else?



Zero Shot Learning

No further training of the model
on a task

Fine Tuning LLMs

Further training of the task or
domain specific data

Using Large Language Models

Zero-Shot Learning — Deep Dive



Definition

Zero-shot = using the pre-trained LLM with no task-specific examples or parameter updates. The model applies its general knowledge to a new task purely through the prompt instruction.

When to use Zero-Shot:

-  Task is common and well-represented in training data (summarization, translation, classification)
-  You need a quick prototype or don't have labeled examples
-  Using a frontier model (GPT-4, Claude 3.5, Gemini 1.5) — they generalize extremely well
-  Avoid for highly specialized domains (medical coding, proprietary jargon, niche legal formats)
-  Avoid when consistent structured output format is critical in production

Supervised Fine-Tuning (SFT) — Deep Dive

Fine-tuning updates the model's weights on a curated dataset of (prompt, ideal_response) pairs for your specific task.

1

Collect Data

Curate 1K–100K (prompt, response) pairs. Quality > quantity — noisy data hurts.

2

Choose Base Model

LLaMA 3.1, Mistral, GPT-3.5-turbo. Larger base = better ceiling but more cost.

3

Configure Training

Learning rate, epochs, batch size. Use PEFT/LoRA to reduce compute 10–100×.

4

Evaluate

Benchmark on held-out test set. Compare to zero-shot baseline. Check for catastrophic forgetting.

5

Deploy

Serve via vLLM / TGI / API. Monitor drift and retrain periodically.

LoRA & PEFT — Efficient Fine-Tuning



Why PEFT?

Full fine-tuning a 70B model requires 8× A100 80GB GPUs for days (\$5K+). PEFT (Parameter-Efficient Fine-Tuning) methods train a tiny fraction of parameters, achieving 95%+ of full fine-tune quality at 1/100th the cost.



LoRA (Low-Rank Adaptation)

Instead of updating full weight matrix W , train two small matrices A and B where $W' = W + AB$. Typically 0.1–1% of total params trained. Works on any linear layer. Merge at inference = zero latency overhead.



QLoRA

LoRA + quantization: load the base model in 4-bit NF4 format, saving 4–8× memory. Fine-tune a 70B model on a single A100 GPU. Benchmark: within 1–2% of full fine-tune. Key paper: Dettmers et al., 2023.



Real-World Cost Comparison

Full fine-tune 7B model: ~\$200, 8 hours on 4× A100 | QLoRA 7B: ~\$8, 2 hours on 1× A100 | QLoRA 70B: ~\$50, 4 hours on 1× A100